COHERENCE OF A PUSHFORWARD TESTED ON THE INTEGRAL SUBSCHEMES THROUGH THE ASSOCIATED POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-local-cohomology/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The associated points of a coherent module and their finiteness on an affine open. Invoked inside the counted proof at source lines 1656-1663 and attributed to lemmas of another chapter (cross-chapter pointers). Make the induction on the number of associated points possible.
- Standard exact sequences and hom descriptions for coherent modules on Noetherian schemes. Several lemmas of the coherent chapter are invoked in the course of the proof (cross-chapter pointers). Provide the local description of the pushforward and the injectivity checks used in the induction step.

Source reference. Original reference algebra-lemma-support-point: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1667–1667. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-support-point, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference coherent-lemma-coherent-Noetherian-quasi-coherent-sub-quotient: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1702–1702. Navigation pointer only: the counted text refers to the Stacks item coherent-lemma-coherent-Noetherian-quasi-coherent-sub-quotient, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:local-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference coherent-lemma-coherent-abelian-Noetherian: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1704–1704. Navigation pointer only: the counted text refers to the Stacks item coherent-lemma-coherent-abelian-Noetherian, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference coherent-lemma-homs-over-open: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1680–1680. Navigation pointer only: the counted text refers to the Stacks item coherent-lemma-homs-over-open, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference coherent-lemma-map-stalks-local-map: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1675–1675. Navigation pointer only: the counted text refers to the Stacks item coherent-lemma-map-stalks-local-map, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference divisors-lemma-check-injective-on-ass: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1717–1717. Navigation pointer only: the counted text refers to the Stacks item divisors-lemma-check-injective-on-ass, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference divisors-lemma-finite-ass: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1660–1660. Navigation pointer only: the counted text refers to the Stacks item divisors-lemma-finite-ass, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference divisors-lemma-ses-ass: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1687–1687. Navigation pointer only: the counted text refers to the Stacks item divisors-lemma-ses-ass, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference schemes-lemma-push-forward-quasi-coherent: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1698–1698. Navigation pointer only: the counted text refers to the Stacks item schemes-lemma-push-forward-quasi-coherent, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let X be a locally Noetherian scheme. Let $j : U \to X$ be the inclusion of an open subscheme with complement Z. For $x \in U$ let $i_x : W_x \to U$ be*

the integral closed subscheme with generic point x. Let $\mathcal{F}$ be a coherent $\mathcal{O}_U$-module. The following are equivalent

(1) *for all $x \in Ass(\mathcal{F})$ the $\mathcal{O}_X$-module $j_*i_{x,*}\mathcal{O}_{W_x}$ is coherent,*
(2) *$j_*\mathcal{F}$ is coherent.*

Proof. We first prove that (1) implies (2). Assume (1) holds. The statement is local on X, hence we may assume X is affine. Then U is quasi-compact, hence $\mathrm{Ass}(\mathcal{F})$ is finite (Divisors, Lemma divisors-lemma-finite-ass). Thus we may argue by induction on the number of associated points. Let $x \in U$ be a generic point of an irreducible component of the support of $\mathcal{F}$. By Divisors, Lemma divisors-lemma-finite-ass we have $x \in \mathrm{Ass}(\mathcal{F})$. By our choice of x we have $\dim(\mathcal{F}_x) = 0$ as $\mathcal{O}_{X,x}$-module. Hence $\mathcal{F}_x$ has finite length as an $\mathcal{O}_{X,x}$-module (Algebra, Lemma algebra-lemma-support-point). Thus we may use induction on this length.

Set $\mathcal{G} = j_*i_{x,*}\mathcal{O}_{W_x}$. This is a coherent $\mathcal{O}_X$-module by assumption. We have $\mathcal{G}_x = \kappa(x)$. Choose a nonzero map $\varphi_x : \mathcal{F}_x \to \kappa(x) = \mathcal{G}_x$. By Cohomology of Schemes, Lemma coherent-lemma-map-stalks-local-map there is an open $x \in V \subset U$ and a map $\varphi_V : \mathcal{F}|_V \to \mathcal{G}|_V$ whose stalk at x is φ_x. Choose $f \in \Gamma(X, \mathcal{O}_X)$ which does not vanish at x such that $D(f) \subset V$. By Cohomology of Schemes, Lemma coherent-lemma-homs-over-open (for example) we see that φ_V extends to $f^n\mathcal{F} \to \mathcal{G}|_U$ for some n. Precomposing with multiplication by f^n we obtain a map $\mathcal{F} \to \mathcal{G}|_U$ whose stalk at x is nonzero. Let $\mathcal{F}' \subset \mathcal{F}$ be the kernel. Note that $\mathrm{Ass}(\mathcal{F}') \subset \mathrm{Ass}(\mathcal{F})$, see Divisors, Lemma divisors-lemma-ses-ass. Since $\mathrm{length}_{\mathcal{O}_{X,x}}(\mathcal{F}'_x) = \mathrm{length}_{\mathcal{O}_{X,x}}(\mathcal{F}_x) - 1$ we may apply the induction hypothesis to conclude $j_*\mathcal{F}'$ is coherent. Since $\mathcal{G} = j_*(\mathcal{G}|_U) = j_*i_{x,*}\mathcal{O}_{W_x}$ is coherent, we can consider the exact sequence

$$0 \to j_*\mathcal{F}' \to j_*\mathcal{F} \to \mathcal{G}$$

By Schemes, Lemma schemes-lemma-push-forward-quasi-coherent the sheaf $j_*\mathcal{F}$ is quasi-coherent. Hence the image of $j_*\mathcal{F}$ in $j_*(\mathcal{G}|_U)$ is coherent by Cohomology of Schemes, Lemma coherent-lemma-coherent-Noetherian-quasi-coherent-subquotient. Finally, $j_*\mathcal{F}$ is coherent by Cohomology of Schemes, Lemma coherent-lemma-coherent-abelian-Noetherian.

Assume (2) holds. Exactly in the same manner as above we reduce to the case X affine. We pick $x \in \mathrm{Ass}(\mathcal{F})$ and we set $\mathcal{G} = j_*i_{x,*}\mathcal{O}_{W_x}$. Then we choose a nonzero map $\varphi_x : \mathcal{G}_x = \kappa(x) \to \mathcal{F}_x$ which exists exactly because x is an associated point of $\mathcal{F}$. Arguing exactly as above we may assume φ_x extends to an $\mathcal{O}_U$-module map $\varphi : \mathcal{G}|_U \to \mathcal{F}$. Then φ is injective (for example by Divisors, Lemma divisors-lemma-check-injective-on-ass) and we find an injective map $\mathcal{G} = j_*(\mathcal{G}|_V) \to j_*\mathcal{F}$. Thus (1) holds. □